Vigneshwari Jayaprakash

Data Scientist · LLM Applications · Production ML

505-550-9282 | vjayapr1@asu.edu | LinkedIn | GitHub | Portfolio

Education

Arizona State University — Tempe, AZ
*M.S. in Data Science (Computing & Decision Analytics) — GPA: 4.0/4.0 | **ASU Scholar*** — *Aug 2024 – May 2026*

Anna University — India
*B.Tech in Information Technology — GPA: 3.6/4.0 | **Gold Medalist*** — *2009 – 2013*

Technical Skills

Languages: Python, SQL, R
ML & AI: Scikit-learn, XGBoost, Random Forest, PyTorch, TensorFlow, LangChain, LangGraph, RAG Systems
MLOps: MLflow, PySpark, Airflow, FastAPI, Docker, Kubernetes, AWS, CI/CD
Statistics: A/B Testing, Hypothesis Testing, Causal Inference, Time-Series Forecasting, Feature Engineering
Viz & BI: Tableau, Power BI, Streamlit

Professional Experience

New Mexico Department of Information Technology — NM, USA
Data Scientist Intern — *Jun 2025 – Dec 2025*

- Built a **LangGraph + RAG platform** on AWS Lambda & S3 that processed **10K+ cybersecurity incidents**, cutting analyst triage time by **70%** through multi-step retrieval and autonomous summarization.
- Reduced **query latency by 40%** by profiling embedding bottlenecks and adding batching with semantic caching; deployed an anomaly detection model via **FastAPI** achieving **85% accuracy** with **60% fewer false positives**.

Infosys Ltd, Client: BNSF Railway — India
Technology Analyst — *Jan 2017 – Jan 2019*

- Built an end-to-end ML fraud detection system processing **1M+ daily transactions** at sub-200ms latency, delivering **$1M+ in annual savings** and earning a formal **CFO commendation**.
- Ran A/B tests and causal inference across 5 model iterations, improving **F1 by 18%**; added a SHAP explainability layer that cut stakeholder review cycles by **30%**.

Data Scientist — *Jun 2015 – Jan 2017*

- Deployed predictive maintenance models for **500+ locomotives**, cutting downtime by **25%** and saving **$400K annually**, backed by a PySpark ETL pipeline at **99.9% uptime**.
- Standardized experiment tracking with **MLflow**, cutting deployment time by **35%** and enforcing model versioning across a 6-person engineering team.

Software Engineer — *Oct 2013 – May 2015*

- Built ML data pipelines processing **500K+ daily records** at 99%+ completeness, with automated validation frameworks that reduced defect rates by **45%** across 3 production systems.

AI & Machine Learning Research (Independent) — United States
Self-Directed Specialization — *2019 – 2024*

- Specialized in **transformer architectures, LLM fine-tuning, RAG systems**, and cloud-native ML deployment using PyTorch, AWS, and Docker in preparation for graduate study at ASU.

Projects

SentinelRAG: LLM Compliance Screening System | *Python, LangChain, ChromaDB, OpenAI API, FastAPI, Streamlit*

- Built a hybrid RAG pipeline screening **18,700+ entities** via keyword, vector, and knowledge graph search, achieving **99% recall and 100% precision** with a deterministic verdict engine.
- Deployed via FastAPI with a bulk-screening dashboard and tamper-proof audit log, applying **prompt engineering** to handle edge-case entity variations reliably.

AI-Powered Railway Track Inspection | *Python, PyTorch, YOLO, OpenCV, Docker, Streamlit*

- Built a deep learning inference pipeline using YOLO on **500+ hours of footage**, achieving **92% defect detection accuracy** and reducing manual inspection effort by **40%**.
- Containerized the inference stack with **Docker** and deployed a real-time Streamlit dashboard with confidence-based filtering, cutting manual review time by **35%**.

Awards & Recognition

CFO Recognition — BNSF Railway: $1M+ annual savings via ML-driven fraud detection
Insta Award — Infosys: recognized for legacy system modernization and engineering impact